

IoT-based energy management for retailers selling household appliances, aligning with EU energy labelling compliance.

Retail × Energy optimisation × IoT platform · OS051 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	50 can we play, can we win	HIGH 0 named in the evidence	€1.0bn modelled evidence · per year	NOW budgeted procurement within 1...	L0 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity helps retailers selling household appliances turn EU energy labelling compliance into a customer-facing energy management service. By using IoT data from connected appliances, retailers can offer personalised energy insights, aligning with new EU labelling rules for washing machines and dishwashers. The timing is now because new EU regulations are in force and retailers need to adapt their sales and after-sales processes.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.9bn €557m – €6.9bn	€1.0bn €300m – €3.7bn	€15.7m €4.5m – €55.8m	modelled
Observed: tendered contract value, annualised	€39.1m	€39.1m	€586k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Retail (EU27_2020)	195,962 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting IoT platform	29.5 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	20,550 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

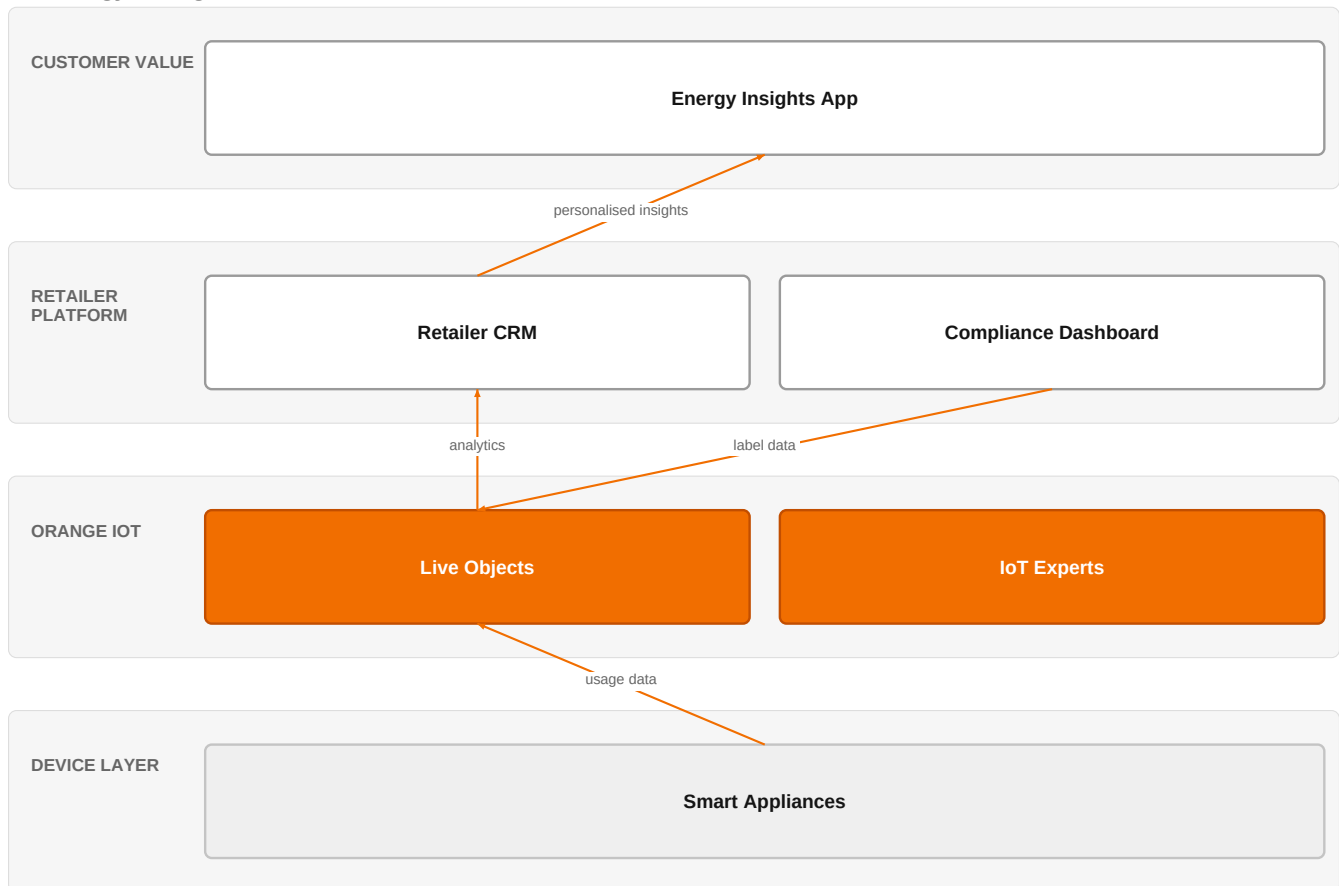
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€39.1m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-20..2026-06-25
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on use case via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

IoT Energy Management for Retailers



■ Orange asset
 ■ Partner
 ■ Customer-owned
 ■ Third party / to be sourced

This diagram shows how Orange's Live Objects platform connects smart appliances to a retailer's CRM, enabling both compliance dashboards and customer-facing energy insights.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU has updated energy labelling requirements for household washing machines and dishwashers, forcing retailers to display new labels and provide accurate energy consumption information at the point of sale. This regulatory change is pushing retailers to rethink how they present energy data and engage customers post-purchase. Meanwhile, IoT platforms and digital twin technologies are maturing, enabling real-time energy monitoring and optimisation that can be offered as a value-added service.

Sources: SIG-DB3990C67390 ec.europa.eu 2026-04-27; SIG-8F0739E8B909 ec.europa.eu 2026-04-23; SIG-216688A4B953 openalex.org 2026-03-01

Who buys, and why now

The buyer is typically the head of digital transformation or the chief commercial officer at a large appliance retailer or a retail group with a significant appliance category. The pain is felt by store operations and after-sales teams who must manage new label displays and answer customer questions about energy consumption. The trigger is the upcoming enforcement of the new labelling rules, which creates urgency to update systems and potentially differentiate through energy services.

Why it is hot now — every claim bound to a dated source

- EU energy labelling requirements for household washing machines and dishwashers affect retail display and sales information. [SIG-DB3990C67390, SIG-8F0739E8B909]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Live Objects IoT platform to connect and manage data from smart appliances (with customer consent), and its IoT experts to design and deploy the solution. The engagement would start with a discovery workshop to map the retailer's current labelling and data infrastructure, then deploy Live Objects to collect energy usage data from appliances, and finally build a customer-facing dashboard or app that provides personalised energy insights and recommendations, leveraging Orange Group researchers for analytics.

Why Orange can win

Orange can win because it has a proven reference with Boulanger, a major French electronics retailer, demonstrating its ability to deliver IoT-based solutions in the retail sector. The combination of Live Objects as a scalable IoT platform and the expertise of Orange Group researchers provides a credible, end-to-end offering. However, the assets are relatively thin compared to hyperscalers, so Orange must emphasise its local presence and ability to integrate with existing retail systems.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	LO	offer addresses the use case AND provides the technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Boulanger	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells IoT platform; competes in OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; active in Retail	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; active in Retail; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently preparing your stores and online channels for the new EU energy labels on washing machines and dishwashers?
- 2 Do you have any existing IoT or smart appliance initiatives, and if so, what data are you collecting today?
- 3 What is your current approach to post-purchase customer engagement, and would you consider offering energy insights as a value-added service?
- 4 Who in your organisation owns the energy labelling compliance project, and what is their budget and timeline?
- 5 Have you explored partnerships with appliance manufacturers to access usage data, and what were the outcomes?
- 6 What are your key criteria for selecting an IoT platform partner, and how do you weigh data sovereignty and security?

Objections you will meet

They will say	You can answer
We already have a compliance team handling labels; we don't need a new service.	That's fair—compliance is the trigger, but the opportunity is to turn that mandatory change into a customer retention and upsell channel. With Live Objects, you can offer personalised energy reports that increase engagement and loyalty.
We're not sure customers will share their appliance data.	Privacy is a real concern, but with transparent opt-in and clear value (e.g., savings tips), many customers are willing. We can design the consent flow to be simple and build trust, and you can start with a pilot to test adoption.
We already have a relationship with a cloud provider like AWS.	That's fine—AWS is also a partner of ours. We can build on your existing infrastructure, but we bring deep IoT expertise and a proven retail reference with Boulanger, which can accelerate your time-to-value.

Risks and unknowns

A key risk is that retailers may not see the value beyond compliance, especially if they lack a strategy for monetising energy data. There is also uncertainty about customer willingness to share appliance usage data, and about the technical complexity of integrating with a wide range of appliance manufacturers. Additionally, the regulatory landscape may evolve, requiring flexibility in the solution.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: assemble a proposal for a retailer that frames Live Objects as the data backbone for real-time energy-labelling display, leveraging the Boulanger reference and the IoT experts' credibility.
Sales	In a conversation with a European appliance retailer's head of digital transformation, say: 'We've helped Boulanger with IoT-based services; we can apply the same platform to make EU energy-labelling compliance a live, customer-facing advantage.'
Strategist / Innovator	Deep-dive brief: evaluate how Live Objects can underpin an energy-labelling compliance dashboard for appliance retailers, using Boulanger as the reference and the IoT experts and Orange Group researchers as the capability pool.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FC4FCB6FF0CC	2026-08-05	ted.europa.eu	1	Italy – Energy-management services – Gara europea per l'affidamento in concessione, tramite fin...
SIG-FE5F2FB82E96	2026-08-05	ted.europa.eu	1	Germany – Electrical installation work of heating and other electrical building-equipment – Ene...
SIG-92171B012A3A	2026-07-22	ted.europa.eu	1	Spain – Energy-efficiency consultancy services – Servicios para la gestion del programa de prev...
SIG-98F9B9FD3AF6	2026-07-22	ted.europa.eu	1	Portugal – Energy and related services – Contrato de Concessao (concecao, construcao, manutenca...
SIG-A112E6B4A541	2026-07-08	ted.europa.eu	1	France – Energy and related services – 26552 - Accord-cadre d'accompagnateur Thermique pour la ...
SIG-3164F57D73A7	2026-06-10	ted.europa.eu	1	Poland – Electricity distribution and control apparatus – Dostawy zabezpieczen pola dla Stacji ...
SIG-904EDD8843E8	2026-06-10	ted.europa.eu	1	Hungary – Energy-management services – Naperomu es tarolo villamos energia értékesítése
SIG-62C1F49E1CA5	2026-05-27	ted.europa.eu	1	Belgium – Heating engineering services for buildings – Raamovereenkomst: Externe expertise warm...
SIG-C094147C3331	2026-05-27	ted.europa.eu	1	Romania – Feasibility study, advisory service, analysis – Servicii de audit energetic, expertiz...
SIG-07C24AAAC8C8	2026-05-20	ted.europa.eu	1	Ireland – Energy and related services – Research and Policy Insights Framework
SIG-47AABAF1373E	2026-05-20	ted.europa.eu	1	France – Energy-management services – Mission d'exploitation, de suivi des performances et d'am...
SIG-DB3990C67390	2026-04-27	ec.europa.eu	1	Energy labelling requirements for household washing machines and household washer-dryers
SIG-8F0739E8B909	2026-04-23	ec.europa.eu	1	Energy labelling requirements for household dishwashers
SIG-216688A4B953	2026-03-01	openalex.org	1	Digital Twin-Enabled Thermal Energy Management System for Sustainable Manufacturing Process Opt...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:30+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.